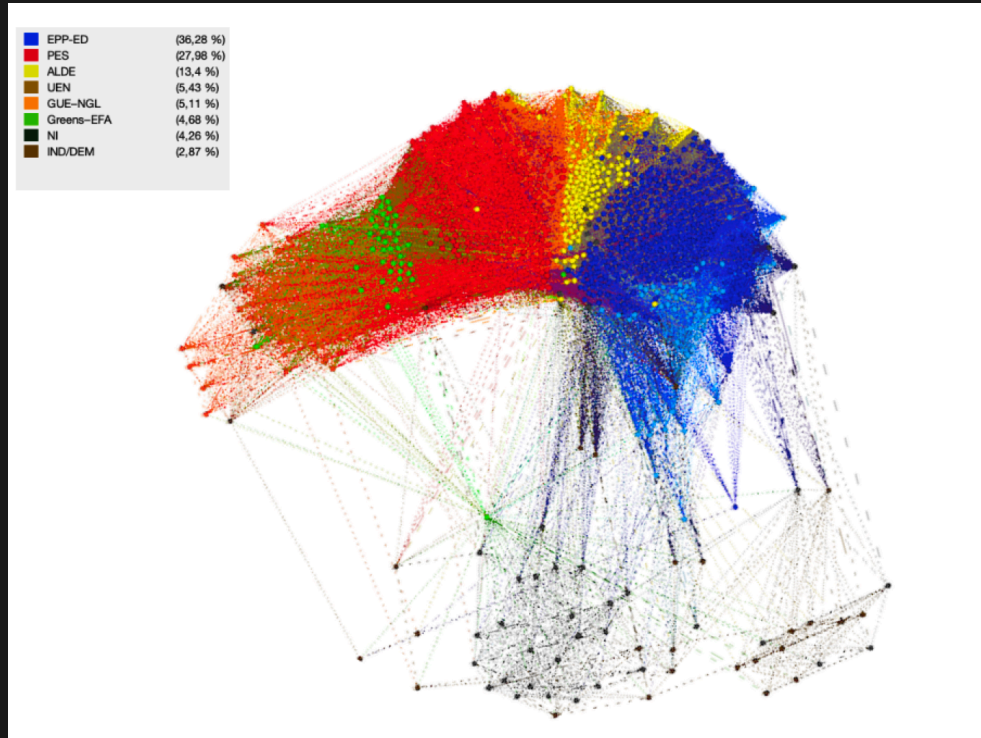
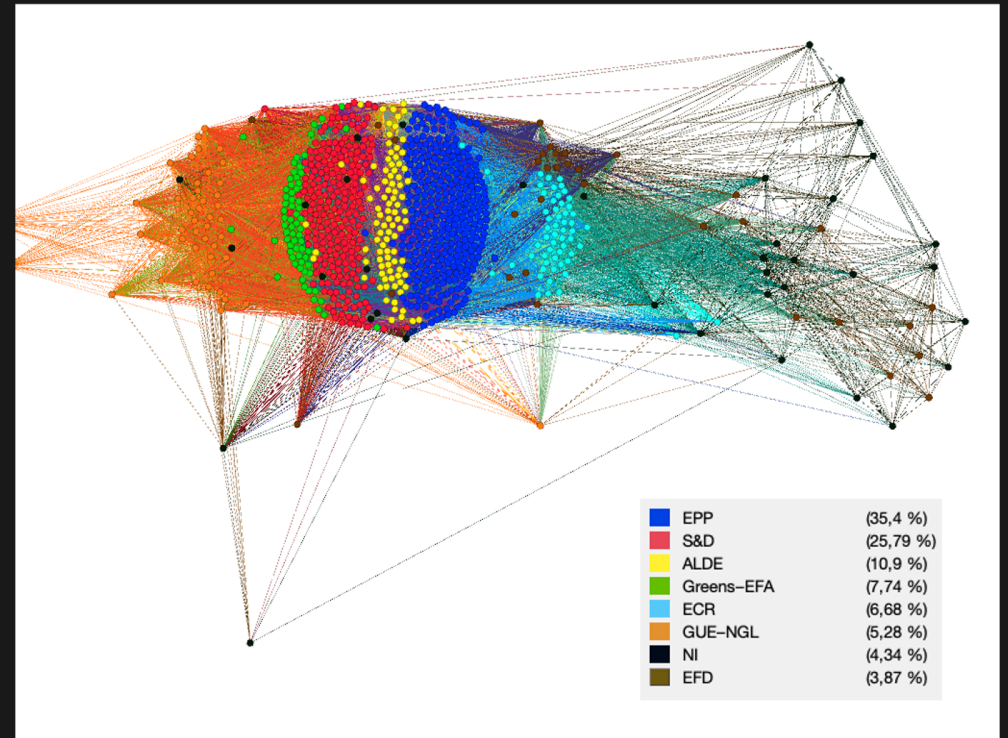


- EP6



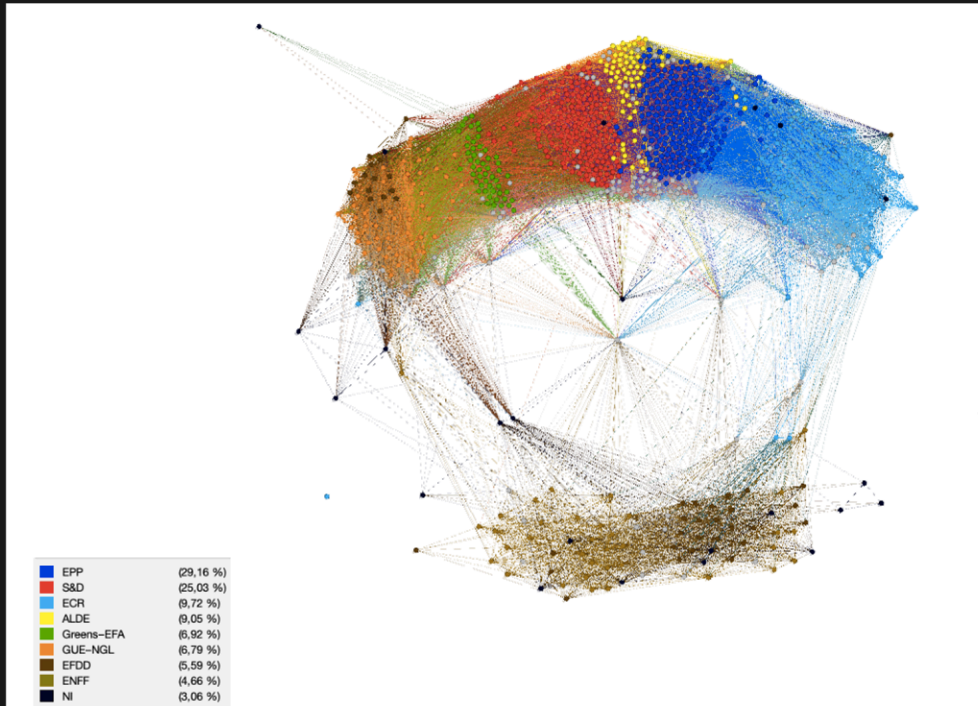
- IND/DEM: UKIP, Lega Nord

- EP7



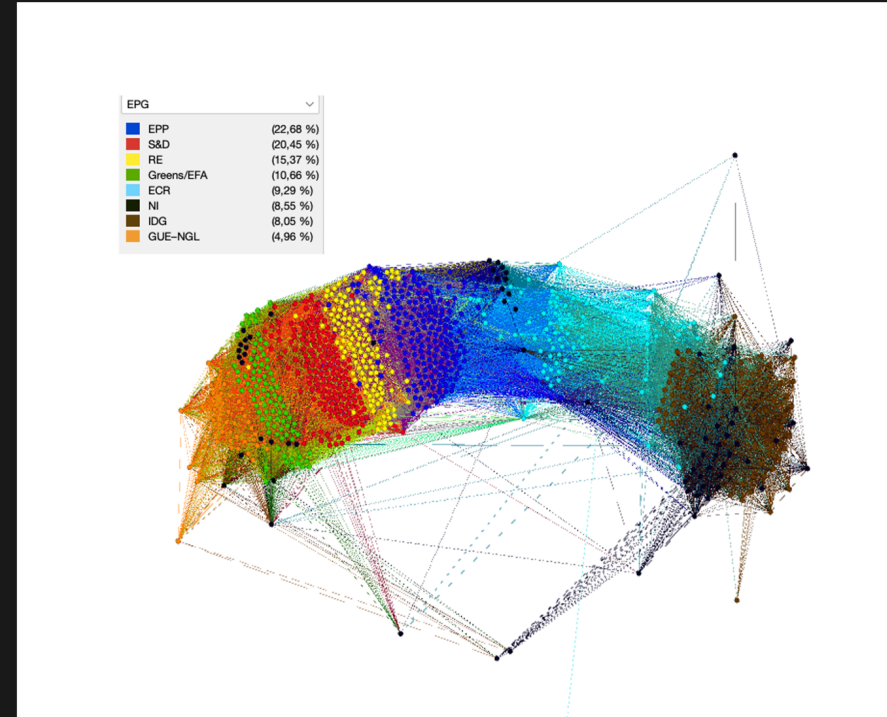
- ECR: Torries; EFD: UKIP, Lega Nord

- EP8



- ECR: Tories, AfD, EFD:
Brexit, Five Star; ENF: Front
National, Lega Nord

- EP9



- ID: Lega Nord, Front
National, AfD; ECR: Tories,
Brothers of Italy, Vox

Two-mode analysis

- The one-mode visualizations are helpful for exploratory analysis
- However, what is needed is a structural modeling approach that works with the whole dataset (as opposed to its 1-d marginalization) and one that avoids rather arbitrary decisions in the construction of the co-voting network
- We will do this by extending appropriately the framework we have developed thus far - effectively, moving from linear to bilinear predictors

From varying coefficients to random factors

- For the moment, let's focus on the simplest data structure within the motivating application
 - i stands for a voting event (some MEP voting for some vote)
 - y_i is 0/1 (“no”/“yes”) - in fact there is absention too, but we discuss this later
 - there are two categorical covariates: MEP and vt

- Along the lines of what we did earlier we associate different parameters to the different levels of the two categorical covariates
- These are meant to learn **ideological characteristics** associated to both MEPs and votes
- We allow these to be multidimensional and we model them probabilistically
 - Among the reasons listed earlier for random effect modeling consider: regularization, number of parameters increasing with sample size, dependence

- Therefore, we assume

$$\begin{aligned}\boldsymbol{u}_k^{\text{MEP}} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{MEP}}) \\ \boldsymbol{v}_\ell^{\text{vt}} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{vt}})\end{aligned}$$

- The difference in notation ($\boldsymbol{u}$, $\boldsymbol{v}$ as opposed to $\boldsymbol{\alpha}$) anticipates an important difference on how these are used in the predictor

- As opposed to previous approach, we now build a **bilinear predictor**

$$\eta_i = \left(\mathbf{u}_{k[i]}^{\text{MEP}} \right)^T \mathbf{v}_{\ell[i]}^{\text{vt}}$$

- We call $\mathbf{u}_k^{\text{MEP}}$, $\mathbf{v}_\ell^{\text{vt}}$ **random factors**, as opposed to random coefficients, because rather than multiplying covariate values they multiply each other

- The implication of the model structure is that if $\mathbf{u}_k^{\text{MEP}}$ and $\mathbf{v}_\ell^{\text{vt}}$ are aligned with each other, which mathematically translates to a large positive inner product, then the outcome is likely to be “yes” - and in an analogous manner when they are orthogonal to each other or point to opposite directions
- In that sense the random factors capture a political ideological identity
- This is an example of an **item response theory (IRT) model**

- Notice that if one of the two factors were actually observed, then the model collapses to the **random slopes model** and the predictor is linear
- For example, this is adopted in ¹ in the context of **recommender systems**
 - This is an application very different in nature from the one considered here but one that is commonly approached using the same type of models (we return to these connections later)

- In the article the user preferences are modeled using random coefficients and the movie characteristics are chosen to be some precomputed functions that relate to genre and other metadata
- However, there are more connections between random factors and coefficients and we discuss them further later on

- The standard IRT model in political science for analyzing roll-call data, see e.g., ¹ combines varying intercepts and random factors:

$$\eta_i = \beta_{\ell[i]} + u_{k[i]}^{\text{MEP}} v_{\ell[i]}^{\text{vt}}$$

- where for a number of reasons that have to do with:
 - legacy decisions
 - regularization arguments
 - computational constraints
 - the american political reality that this is commonly applied to
- the varying intercepts are treated as fixed effects
- the random factors are one-dimensional

- In this particular context the random factors for legislator are known as **ideal points** and those for votes as **discrimination parameters**
- The results below are obtained from **voteview** and the estimation is using what is known as **nominate**, which is an optimization-based (and little primitive) software



RUBIO, Marco (1971-)

Republican of Florida 

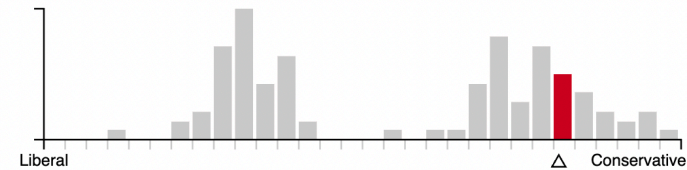
Serving in Senate 2011-Present

119th Congress (2025-2027) ▾

[View all members](#)

Ideology

[Attendance and Loyalty](#)



CRUZ, Rafael Edward (Ted) (1970-)

Republican of Texas 

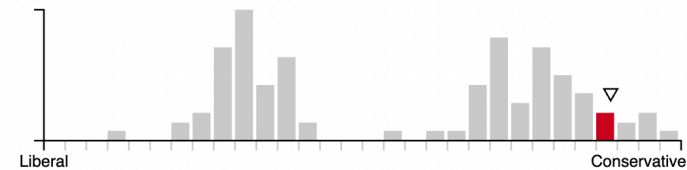
Serving in Senate 2013-Present

119th Congress (2025-2027) ▾

[View all members](#)

Ideology

[Attendance and Loyalty](#)



- Below we have an illustration from EP using two-dimensional ideal points but also their hierarchical decomposition into an **EPG** term and an idiosyncratic for each MEP
 - It is an example of what is called a **hierarchical IRT model**
- We also add varying intercepts for MEPs and treat all varying coefficients as random

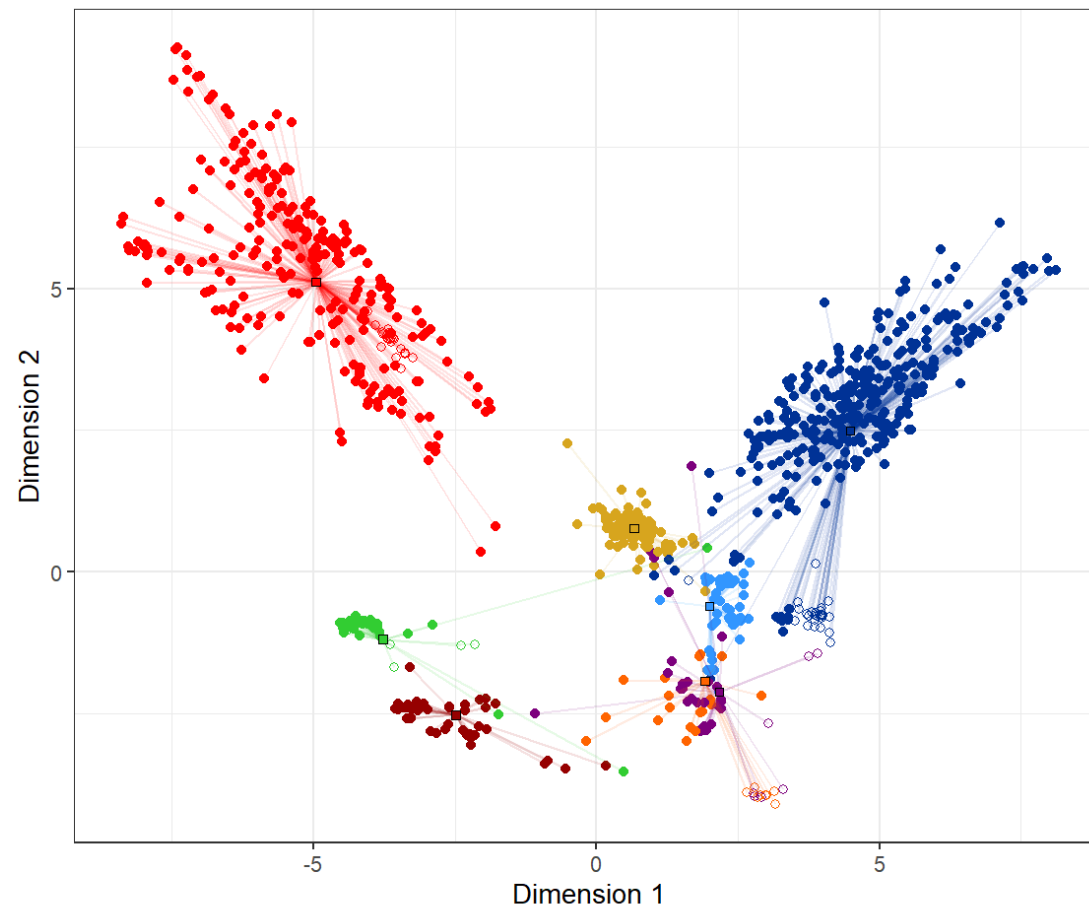
$$\eta_i = \alpha_{k[i]}^{\text{MEP}} + \alpha_{\ell[i]}^{\text{vt}} + \left(\mathbf{u}_{k[i]}^{\text{MEP}} + \mathbf{u}_{j[i]}^{\text{EPG}} \right)^T \mathbf{v}_{\ell[i]}^{\text{vt}}$$

- Observations:
 - **Embedding** on the same **latent space** of person and party ideology
 - The different role of random intercepts and random factors

- Below we show results for EP6 (2004-2009)
 - Our models involves a total of $\mathcal{O}(10^5)$ random effects and are trained on $\mathcal{O}(10^6)$ number of observations
- The model is estimated using VI and our **vglmer** with the model formula:

**(1 | vote_id+MEP_id) + mi(MEP_id ~ EPG,
vote_id,2)**

- We show VI approximation to posterior means



A matrix factorization perspective

- Suppose for simplicity that we were modeling directly the mean of the observed data (i.e., identity link, as opposed to the logistic link used earlier)
- Suppose further that the data is organized in a matrix (e.g., the roll-call matrix), say $\mathbf{Y}$, with elements $Y_{k,\ell}$
- Finally, assume a basic random factor model that assigns random factors $\mathbf{u}_k$ to each row and $\mathbf{v}_\ell$ to each column

- Then, according to the model:

$$\mathbb{E}[Y_{k,\ell} | \mathbf{u}_k, \mathbf{v}_\ell] = \mathbf{u}_k^T \mathbf{v}_\ell$$

- If we were further to create a matrix $\mathbf{U}$, the k 'th row of which is $\mathbf{u}_k^T$, and $\mathbf{V}$, the ℓ 'th row of which is $\mathbf{v}_\ell^T$, then according to the model:

$$\mathbb{E}[\mathbf{Y} | \mathbf{U}, \mathbf{V}] = \mathbf{U}\mathbf{V}^T$$

- Therefore, the random factor model performs a **low-rank decomposition** of the (conditional) mean of the data
- In other words, it performs a **matrix factorization** (the defining characteristic of which is that it is low rank)
- The approach that models the factors as random is known as **probabilistic matrix factorization**, e.g. ¹
- We will discuss this angle, and its links to variational inference, further later

- However, this low-rank approximation perspective offers an interesting link between random coefficients and random factors
- Given two categorical covariates one option is to add to the predictor their **full interaction**, e.g: **(1 | MEP:vt)**
 - This would add a linear term in the predictor but for covariates with many levels this adds lots of random coefficients
 - Additionally, in situations like the roll-call data (or recommendation, to that same effect) we typically observe at most once each cell in the contingency table, hence we need to regularize more

- An alternative is to create a **multiplicative interaction** between the two
 - This is effectively a **low-rank representation** of the full interaction
 - Therefore, we introduce random factors associated to the levels of the two and add their product to the predictor
 - The resulting model has much less latent variables but it becomes non-linear (specifically, bilinear)
- Hence the tension between linear and bilinear terms is one of **statistical vs computational efficiency**

Principal components, factor models, loadings and random factors

- It should already start becoming clear that the multiplicative interaction has close links to both **principal component analysis** and to so-called **factor models**, which we discuss in detail later
- Here we only point out that the fixed vs random effect issue has its analogue also in this context

- It is common, especially in some social science applications, to treat the row effects as random effects (assuming a data organization where the rows outnumber the columns) and the columns as fixed effects
- In such applications the rows are many and the columns are few

- Again, some of the reasons discussed earlier apply here too. We highlight two:
 - Regularization is used for row effects for which there is relatively little information
 - The columns define fixed quantities, e.g., questions (aka items) in a questionnaire, whereas rows respondents that can be seen as drawn from an infinite population

- In these context the matrix of fixed effects $\mathbf{V}$ is known as the **loadings matrix**
- Quite confusingly, some times the columns of $\mathbf{V}$ (or a scaled version thereof) are referred to as **principal components**, e.g. Section 3 in ¹, but others the rows of $\mathbf{U}$ (or a scaled version thereof), e.g., ²

From fixed effects, to random effects to random factor - in a picture!

